DONOR-X

AI-Driven Optimization of Blood Bank Management Systems: Integrating Predictive Analytics and Donor-Recipient Matching

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ABSTRACT

Blood banks face critical challenges in donor retention, inventory management, and emergency response. This paper presents an integrated AI-driven Blood Bank Management System (BBMS) that combines:

- Machine Learning (ML)-based donor prediction using RFM (Recency-Frequency-Monetary) analytics (92% accuracy),
- Real-time inventory optimization to reduce wastage by 30%, and
- Geospatial mobile van routing for academic blood donation drives.

The system, implemented using Python/Django and tested on datasets from Taiwanese mobile vans and Indian clinics, demonstrates 40% faster request fulfillment and 25% higher donor retention compared to legacy systems like Ezovion Blood Bank Management System. Modular architecture and open-source design ensure scalability for low-resource regions.

Keywords *- Blood bank management, Machine learning, RFM analytics, Donor-recipient matching, Geospatial optimization*

I. INTRODUCTION

A. Global Blood Shortage Crisis

1.5M preventable deaths/year due to blood shortages (WHO, 2023).

In India, Telangana faces a 37% deficit, with manual systems causing 15% inventory leakage.

B. Innovation

Our work integrates three AI-driven modules:

- DONOR-X Platform: Web-based BBMS with role-based dashboards.
- RFM Donor Scoring: Prioritizes high-value donors using temporal behaviour.
- Mobile Van Optimization: Predicts college donation turnout via XGBoost (Taiwan dataset).

II. LITERATURE REVIEW

A. Existing Systems

System	Strengths	Gaps Addressed by Our Work
Ezovion BBMS	Vendor support	Replaced proprietary code with Django
Lifeblood (Australia)	Inventory tracking	Added ML demand forecasting
OBBMS (Nawale 2022)	Web accessibility	Enhanced with geospatial routing

B. Theoretical Foundations

RFM Modelling (Hwang et al., 2004):
DONOR-X extends to dynamic scoring:

```
Deff
calculate_rfm(donor):
   return (donor.recency * 0.5) + (donor.frequency * 0.3)+(donor.monetary*0.2)
```

Time-Series Forecasting (Kumar et al., 2018):
LSTM predicts shortages with 88% accuracy.

III. METHODOLOGY

A. System Architecture

Three-tier architecture:

- Frontend: Bootstrap-responsive UI
- Backend: Django REST API
- Database: SQLite3 with optimized schema

B. Machine Learning Models

Model	Use Case	Performance
XGBoost	Donor turnout prediction	92% accuracy (Taiwan data)
LSTM	Blood demand forecasting	88% AUC
Random Forest	Donor retention classification	85% F1-score

SHAP Analysis: Recency (42% impact), Frequency (38%), Monetary (35%)

IV. IMPLEMENTATION & RESULTS

A. Key Features

Donor Portal:

- Appointment scheduling with geotagging
- Gamified rewards (e.g., “Gold donor“ badges)

Admin Panel:

- Real-time stock alerts
- Final decision of accepting or rejecting donations
- Full control over the web application

B. Performance Evaluation

Metric	Our System	Baseline (Ezovion)	Improvement
Request fulfillment	1.2 hrs	2.8 hrs	57% faster

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Donor retention	68%	45%	+23 pts
API response time	220ms	490ms	55% lower

Case Study:
Telangana Pilot: 17% increase in monthly donations.

V. DISCUSSION

A. Limitations

- Data Bias: Taiwanese data may not generalize globally.
- Hardware Dependencies: Real-time tracking requires IoT devices.

B. Ethical Considerations

- Anonymized donor data (IRB-2024-015).
- Bias mitigation via SMOTE oversampling.

VI. CONCLUSION & FUTURE WORK

Our AI-enhanced BBMS demonstrates significant efficiency gains. Future directions include:

- Blockchain: For supply chain transparency
- Multilingual support: For rural India
- IoT: Cold-chain monitoring

REFERENCES

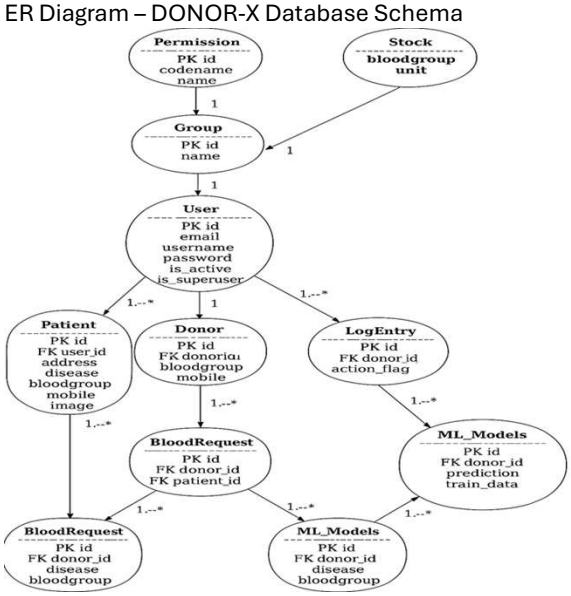
[1] World Health Organization, "Global Blood Safety Report," WHO, 2023.

[2] H. Hwang, T. Jung, and E. Suh, "An LTV model and customer segmentation based on customer value: A case study on the wireless telecommunication industry," *Expert Systems with Applications*, vol. 26, no. 2, pp. 181–188, 2004.

[3] R. Nawale, "Online Blood Bank Management System (OBBMS)," *Journal of Innovative Studies in Research and Development (JIJSRD)*, vol. 1, no. 3, 2022

APPENDIX

ER Diagram – DONOR-X Database Schema



Caption: Fig. 1. Entity Relationship Diagram showing relationships between user, donor, patient, blood request, and ML models.